| Risk Assessment No: WF042 | Task: Welding Fume and Local Exhaust Ventilation (LEV) | |
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| **Prepared By:** Derick Bibby | **Date:** 15.01.2026 | **RA Review Date:** Annually |
| | **Reviewed By:** Lee Hardy | **Date:** 15.01.2026 |

**PPE:** Safety Footwear, High Visibility Jacket, High Visibility Trousers, Flame-retardant welding overalls, Welding gauntlets, Welding helmet with suitable filter shade, Respiratory Protective Equipment (RPE) as specified below – powered air-purifying (PAPR) welding helmet as standard

| Activity | Hazards | S | L | R | Control Measures | L | S | R |
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| Welding indoors in the workshop or fabrication bay (MIG/MAG, MMA/stick, TIG, flux-cored) – fabrication, chassis and body repair | Inhalation of welding fume – all welding fume, including mild steel, is a human carcinogen (lung cancer)<br><br>Manganese in fume – neurological damage<br><br>Gases: ozone, nitrogen oxides, carbon monoxide<br><br>Metal fume fever, asthma, COPD<br><br>Exposure of others working nearby | 5 | 4 | 20 | • **Legal position:** HSE will not accept any welding without suitable exposure control, regardless of duration. General ventilation or open doors on their own are NOT adequate control.<br>• **Reduce first:** consider bolting or other cold-joining methods; use lower-fume processes and consumables (e.g. pulsed MIG, solid wire instead of flux-cored where suitable); keep current/voltage to the lowest effective setting.<br>• **Clean the metal:** remove oil, grease, paint, underseal and galvanising from the weld area before welding (see row 4).<br>• All workshop welding is to be carried out in the designated Hot Works / Fabrication area unless a hot work permit covers another location (see WF005 Hot Works).<br>• **LEV is mandatory for all indoor welding:** use on-torch extraction for MIG/MAG where fitted; otherwise use the extracted bench/booth or a movable capture hood on an extraction arm.<br>• Position the capture hood close to the weld (within the capture distance marked on the hood or stated in the LEV user manual) and to the side, so fume is drawn away from the breathing zone. Reposition the hood as the weld progresses.<br>• Switch LEV on before striking the arc and leave it running until the fume has cleared.<br>• Check the airflow indicator / gauge before use. If the LEV is not working, damaged, or the indicator shows low flow – do not weld. Tag it out and report it to the Workshop Supervisor.<br>• Do not position your head in the fume plume. Arrange the workpiece so you can work to the side of the rising fume.<br>• Mechanical general ventilation (roof/wall extract fans) is to be kept running during welding to clear residual fume.<br>• Use welding screens/curtains (BS EN ISO 25980) and signage to keep others out of the welding area. | 2 | 5 | 10 |
| Residual fume where LEV alone does not give | Inhalation of residual fume that escapes capture | 5 | 3 | 15 | • **RPE is required in addition to LEV** wherever LEV cannot fully control the fume (e.g. large or awkward workpieces, overhead or positional welding, work inside trailer bodies, or where an exposure survey shows residual exposure). | 2 | 5 | 10 |

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| adequate control | Ineffective RPE due to poor fit, facial hair, flat batteries or blocked filters | | | | • **Standard issue for regular welders:** powered air-purifying respirator (PAPR) welding helmet with particle filter. Loose-fitting PAPR does not need a face-fit test and can be worn with facial hair.<br>• **Short, infrequent tasks only:** a tight-fitting half mask with P3 filters or an FFP3 disposable mask (assigned protection factor 20) worn under the welding helmet. Tight-fitting RPE must be face-fit tested to the wearer, and the wearer must be clean shaven where the mask seals.<br>• Before each use: check PAPR battery charge and airflow (using the manufacturer's flow indicator), filter condition and expiry, seals and visor. Do not use defective RPE – report it.<br>• RPE is personal issue. Store it clean, out of the welding area, and replace filters to the manufacturer's schedule.<br>• Fit-test records, RPE issue records and training records are kept by the Workshop Supervisor as part of the RPE programme. | | | |
| Welding outdoors – yard, quarry, site or roadside repairs to wagons, trailers and plant | Inhalation of welding fume – LEV cannot capture fume effectively outdoors<br><br>Wind blowing fume into the welder's or others' breathing zone | 5 | 3 | **15** | • **RPE is mandatory for all outdoor welding** – PAPR welding helmet as standard (or FFP3/P3 for short tasks if face-fit tested). Being outside is NOT a substitute for RPE.<br>• Position yourself upwind of the weld where practicable and keep others away from the fume; use mobile screens.<br>• Site welding to be planned with the site responsible person, under their hot work permit where required (see WF005 and SSoW-047).<br>• This row supersedes the outdoor welding control in WF003 Electrical Welding ("well-ventilated areas"). | 2 | 5 | **10** |
| Welding, cutting or grinding on painted, coated, galvanised, stainless or contaminated metal (wagon bodies, chassis, tanks, trailers) | Isocyanates released from 2-pack (2K) paints when heated – occupational asthma<br><br>Zinc oxide fume from galvanised steel – metal fume fever<br><br>Hexavalent chromium and nickel from stainless steel – cancer, asthma<br><br>Toxic gas (e.g. | 5 | 4 | **20** | • Assume vehicle bodies and chassis are coated with 2K paint unless known otherwise.<br>• Strip paint, underseal and coatings back from the weld/cut area before hot work, using methods that minimise dust (wire wheel or abrasive disc with extraction, or chemical strippers under COSHH assessment – see WF001, WF007 and SSoW-070).<br>• Always use extraction AND RPE when hot working on painted, galvanised or stainless steel. Only trained staff are to carry out hot work on painted panels.<br>• Stainless steel welding and cutting: on-torch or close-capture LEV plus PAPR; keep others out of the area.<br>• Do not use chlorinated solvent degreasers in or near the welding area. Make sure degreased parts are completely dry before welding.<br>• Tanks, drums and containers that have held fuel, oil or other flammable or toxic substances must not be welded or cut until separately risk assessed and cleared under a permit to work (see WF003 and WF005). | 2 | 5 | **10** |

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| | phosgene) from chlorinated degreaser residues exposed to the arc<br><br>Lead from old paint | | | | | | | |
| Thermal cutting and gouging – plasma cutting, oxy-fuel cutting, arc-air gouging | High volumes of metal fume<br><br>Nitrogen oxides from plasma cutting<br><br>Noise (see WF014) | 5 | 4 | **20** | • Use downdraught / extracted cutting tables for plasma cutting of plate in the workshop wherever possible.<br>• For hand-held cutting and gouging, use a movable capture hood or extracted bench plus RPE (PAPR as standard).<br>• Gouging is high-fume – carry out in the designated fabrication area with LEV and RPE, and keep others clear.<br>• Gas cylinder controls as WF002 Compressed Gas Cylinders. | 2 | 5 | **10** |
| Welding in vehicle pits, inside trailer bodies, tanks or other poorly ventilated spaces | Build-up of fume and shielding gases (argon, CO2) which are heavier than air and can collect in pits – asphyxiation<br><br>Oxygen enrichment from leaking oxy-fuel equipment – fire | 5 | 3 | **15** | • Do not weld or use gas equipment in a vehicle pit or enclosed body unless it has been specifically assessed and authorised by the Workshop Supervisor (see WF010 Use of Vehicle Pit and SSoW-054).<br>• Where authorised: use LEV that discharges outside the space, PAPR, gas cylinders kept outside the space, and a second person in attendance.<br>• Never use oxygen to 'freshen the air' or blow down clothing.<br>• Any space that meets the definition of a confined space must not be entered without a confined-space risk assessment and permit. | 1 | 5 | **5** |
| LEV examination, testing and maintenance | LEV deteriorates without anyone noticing – workers exposed while believing they are protected<br><br>Dust and fume exposure during filter changes | 4 | 4 | **16** | • Every LEV unit (fixed, mobile and on-torch) is listed on the workshop LEV register with its user manual, commissioning report and log book.<br>• **Weekly:** visual check of each LEV unit by the user (hood, arm, ducting, filters, airflow indicator) – recorded in the log book.<br>• **At least every 14 months:** statutory thorough examination and test (TExT) by a competent ventilation engineer. Act on the report's recommendations. Keep TExT records for at least 5 years.<br>• Filters are to be changed or cleaned to the manufacturer's schedule. Wear RPE (FFP3 minimum) and disposable gloves, bag used filters and dispose of them as hazardous waste.<br>• Defective LEV is to be tagged "DO NOT USE" and taken out of service until repaired. Welding in that bay stops until suitable alternative extraction is in place. | 2 | 4 | **8** |

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| Health monitoring, training and supervision | Ill health not detected early<br><br>Controls not used correctly | 4 | 3 | **12** | • All welders and fabricators receive instruction and training on welding fume risks, correct use of LEV, and fitting and checking of RPE before starting work. Training is recorded.<br>• Health surveillance (respiratory questionnaire and lung function testing via the occupational health provider) for those who regularly weld, cut or work with stainless steel or 2K-painted metal. Report any breathing symptoms (cough, wheeze, breathlessness) to your supervisor.<br>• Occupational hygiene exposure monitoring to be arranged to confirm that LEV and RPE controls are adequate, and repeated after any significant change in work or equipment.<br>• Supervisors check during routine workshop inspections that LEV is switched on and correctly positioned and RPE is being worn. | 2 | 4 | **8** |

## Risk rating key

S = Severity (1–5), L = Likelihood (1–5), R = Risk rating (S × L). Initial rating shown left of the controls; residual rating (with all controls in place) shown right. Bands follow the handbook risk matrix (page 4): 1–3 Acceptable – low (green); 4–7 Acceptable – moderate (blue); 8–14 Acceptable – constant review (yellow); 15–19 Additional control measures required (red); 20–25 Stop (dark red).

## Related sections

WF002 Compressed Gas Cylinders · WF003 Electrical Welding · WF005 Hot Works · WF007 Dust · WF010 Use of Vehicle Pit · WF014 Noise · WF016 PPE · SSoW-054 Maintenance in Trailer Bodies · SSoW-070 COSHH in the Workshop (draft)

## References


- HSE Safety Alert – Change in enforcement expectations for mild steel welding fume: https://www.hse.gov.uk/safetybulletins/mild-steel-welding-fume.htm
- HSE – Controlling the risks from welding: https://www.hse.gov.uk/welding/welding-controls.htm
- HSE WL3 – Welding fume control: https://www.hse.gov.uk/pubns/guidance/wl3.pdf
- HSE MR6 – COSHH essentials: Hot work on painted panels: https://www.hse.gov.uk/pubns/guidance/mr6.pdf
- HSE – LEV frequently asked questions (14-month TExT): https://www.hse.gov.uk/lev/faqs.htm
- HSE INDG479 – Guidance on RPE fit testing: https://www.hse.gov.uk/pubns/indg479.htm
- Control of Substances Hazardous to Health Regulations 2002 (COSHH), regulations 7, 9 and 11